

```
C:\Users\gaura\Desktop>python dk.py
Database Table created.

Table Records inserted.

Table Record updated.

Table Record deleted.

Remaining records in Table :
```

	id	user	remote_ip	login_time	remote_location	risk_factor_score
0	1	user1	192.168.2.101	2025-06-27 10:15:00	USA	7.8
1	2	user2	202.1.115.42	2025-06-27 11:00:00	Germany	9.5

Figure 4: Output of database operations in DuckDB

```
# Phase 2: Insertion of records and data
duckdbconnection.execute("""
INSERT INTO suspected_logins VALUES
    (1, 'user1', '192.168.2.101', '2025-06-27 10:15:00',
    'USA', 7.8),
    (2, 'user2', '202.1.115.42', '2025-06-27 11:00:00',
    'Germany', 8.9),
    (3, 'user3', '197.52.101.23', '2025-06-27 11:30:00',
    'India', 6.5);
""")
print("Table Records inserted.\n")

# Phase 3: Update values in risk factor score for user2
duckdbconnection.execute("""
UPDATE suspected_logins
SET risk_factor_score = 9.5
WHERE user = 'user2';
""")
print("Table Record updated.\n")

# Phase 4: Delete values in record for user3
duckdbconnection.execute("""
DELETE FROM suspected_logins
WHERE user = 'user3';
```

```
""")
print("Table Record deleted.\n")

# Phase 5: Fetch and display remaining records
result = duckdbconnection.execute("SELECT * FROM suspected_logins").fetchdf()
print("Remaining records in Table :\n", result)
```

With this code, the insert, delete and update operations are implemented in the DuckDB database (Figure 4). High-end applications related to data analytics and data engineering can be executed in a similar format.

Heterogeneous and free-flowing data needs to be controlled and visualised in an effective way with the help of charts, graphs and many other GUI modules. Data scientists need high performance and lightweight database applications to handle such data. DuckDB can work with real-time streaming data, enabling dynamic plotting with multiple programming languages. 

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